Optoelectronics GROUP @ ZJU

浙江大学 ZHEJIANG UNIVERSITY



Fundamentals of Optoelectronics (光电子学基础)

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Greek Letters

Greek Alphabet Symbols List

希腊字母	中文注音	英文注音	国际音标	写法	键盘输入（Symbol	数学含义
Α α	阿尔法	Alpha	/alfa/		A a	角度，系数，角加速度
Β β	贝塔	Beta	/beitə/		B b	磁通系数，角度，系数
Γ γ	伽玛	Gamma	/gamə/		G g	电导系数，角度，比热容比
Δ δ	德尔塔	Delta	/deltə/		D d	变动，密度，屈光度
Ε ε	艾普西龙	Epsilon	/ep'silon/		E e	对数之基数，介电常数
Ζ ζ	捷塔	Zeta	/zi:tə/		Z z	系数，方位角，阻抗，相对粘度
Η η	依塔	Eta	/i:tə/		H h	迟滞系数，效率
Θ θ	西塔	Theta	/θi:tə/		Q q	温度，角度
Ι ι	艾欧塔	Iota	/ai'oute/		I i	微小，一点
Κ κ	卡帕	Kappa	/kapə/		K k	介质常数，绝热指数
Λ λ	拉姆达	Lamda	/lamdə/		L l	波长，体积，导热系数
Μ μ	缪	Mu	/mju:/		M m	磁导系数，微，动摩擦系（因）数，流体粘度

Ν ν	拗	Nu	/nju:/		N n	磁阻系数，动力粘度
Ξ ξ	克西	Xi	/ksai/		X x	随机数，区间内的一个未知特定值
Ο ο	欧麦克轮	Omicron	/oumaik'rən/		O o	高阶无穷小函数
Π π	派	Pi	/pai/		P p v	圆周率，π（n）表示不大于 n 的质数个数
Ρ ρ	柔	Rho	/rou/		R r	电阻系数，柱坐标和极坐标中的极径，密度
Σ σ	西格玛	Sigma	/sigmə/		S s	总和（大），表面密度，跨导，正应力
Τ τ	套	Tau	/tau/		T t	时间常数，切应力
Υ υ	宇普西龙	Upsilon	/ju:p'silən/		U u	位移
Φ φ	服艾	Phi	/fai/		F j	磁通，角，透镜焦度，热流量
Χ χ	器	Chi	/kai/		C c	统计学中有卡方（χ^2）分布
Ψ ψ	普赛	Psi	/psai/		Y y	角速，介质电通量
Ω ω	欧米伽	Omega	/oumigə/		W w	欧姆（大），角速度，交流电的电角度

Quiz: Greek letters

Γ	Gamma	Μ	Mu	Σ	Sigma
Ε	Epsilon	Ν	Nu	Υ	Upsilon
Ι	Iota	Ξ	ξ 克西	Χ	Kai
Λ	Lamda	Ο	Omicron	Ψ	Psi
Η	Eta				

				ξ	克西
γ	Gamma	η	Eta	σ	Sigma
Δ	Delta	ι	Iota	υ	Upsilon
ε	Epsilon	κ	Kappa	χ	Kai
ζ	Zeta	ν	Nu	ψ	Psi

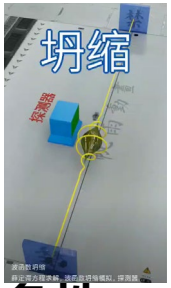
What is light?

17世纪，牛顿力学，光是粒子，解释反射、折射、色散

杨氏双缝干涉，为什么光变暗。惠更斯原理光是波

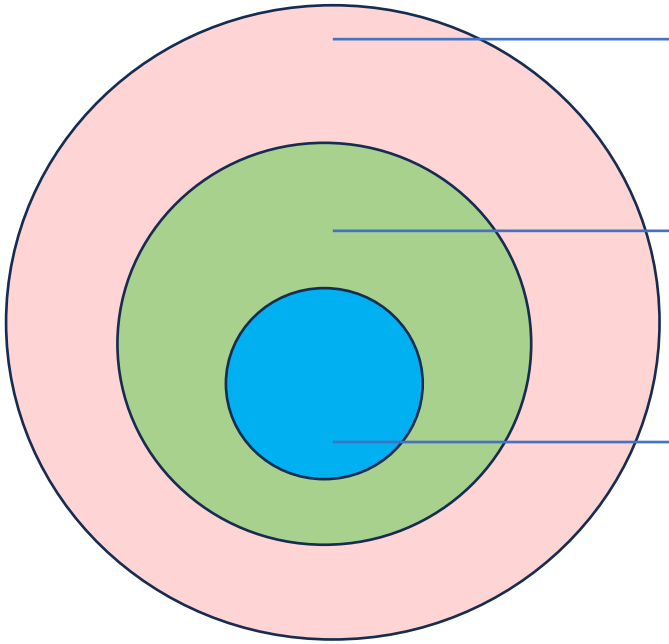
光偏振无法机械波解释，传播介质解释，麦克斯韦推导电磁波速等于光速

光电效应不满足波动理论的光强对应能量，黑体辐射紫外灾难，杰弗里泰勒单光子双缝干涉



几何光学 → **粒子学说** → **波动光学** → **电磁光学** → **波粒二象性**

量子场论： 光被视为电磁场的量子化激发态，表现为离散的能量包，即光子。



Electromagnetic optics: vector theory comprising an electric field and magnetic field that vary in time and space.

1. Our course starts here

Wave optics: approximation to electromagnetic optics that relies on the wavefunction, a scalar function of time and space.

2. Next stage

Ray optics: the limit of wave optics when the wavelength is very short

Maxwell’s Equation in free space

James Clerk Maxwell synthesized the existing knowledge of electricity and magnetism into a coherent theoretical framework. His key contributions include:

(1) Unification of Electric and Magnetic Fields
Maxwell combined the works of Gauss, Faraday, and Ampère into a single set of equations.

(2) Prediction of Electromagnetic Waves
Maxwell's equations predicted that electric and magnetic fields could propagate through space as waves. He calculated the speed of these waves and found it to be equal to the speed of light, leading him to conclude that light is an electromagnetic wave .

$$\nabla \cdot \vec{B} = 0$$

Gauss’s law of magnetism
(高斯磁定律: 磁单极不存在)

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

Gauss’s law of electrostatic
(高斯静电定律：电荷如何产生电场)

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Faraday’s law of of Induction
(法拉第感应定律：时变磁场产生电场)

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Ampere’s Circuital law
(安培定律：时变电场产生磁场)

Maxwell’s Equation 1: Gauss’s law of magnetism

$$\nabla \cdot \vec{B} = 0$$

Divergence (散度）读del或nabla /ˈnæblə/算子，黑体代表矢量算符

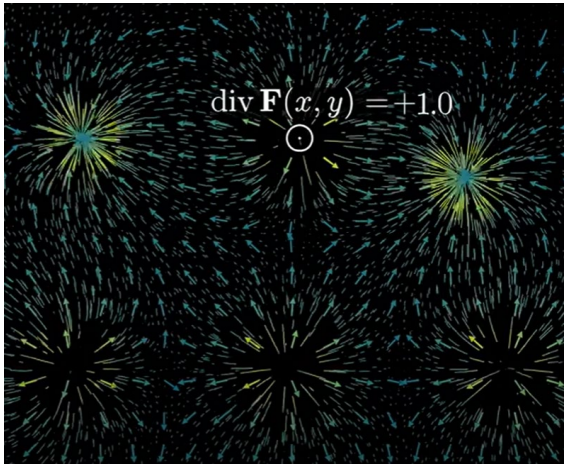
Magnetic flux density

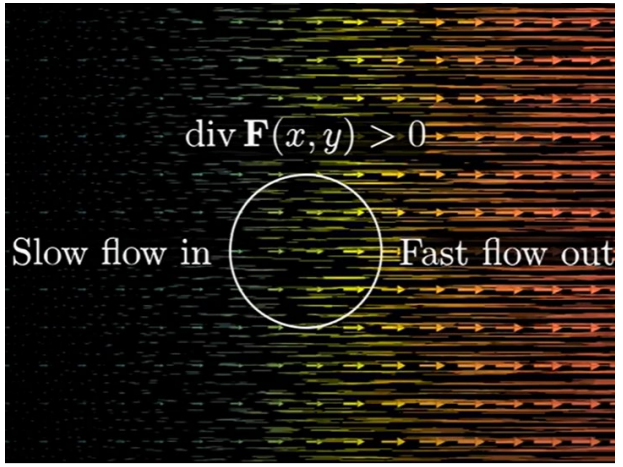
$$\nabla \cdot \vec{B} = \operatorname{div} \vec{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$$ 一阶偏导

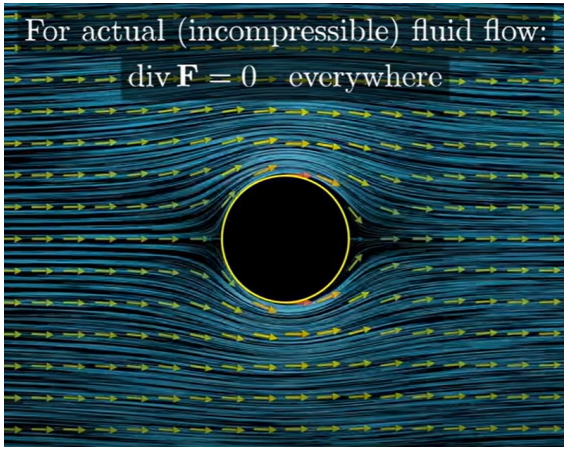
Divergence of a vector field at a particular point of the plane tells you how much the fluid tends to flow out of or into small regions near it.

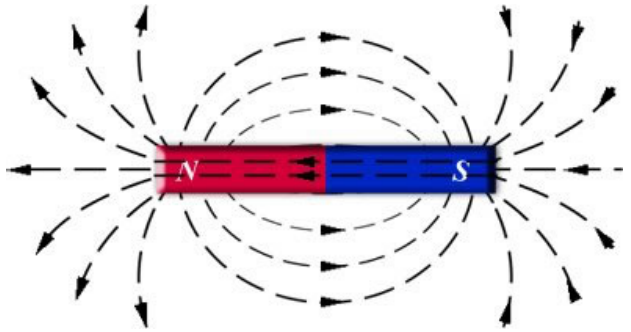
散度（divergence）可用于表征空间各点矢量场发散的强弱程度，能够反映该点通量体积变化率的重要度量。

物理上，散度的意义是场的有源性。当div F>0 ，表示该点有散发通量的正源（发散源）；当div F<0 表示该点有吸收通量的负源（洞或汇）；当div F=0，表示该点的矢量场场线没有发出也没有汇聚。









Maxwell's Equation 1: Gauss's law of magnetism

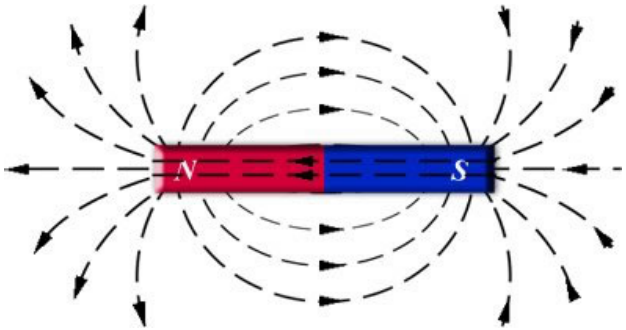
Integral Form | Differential Form

$$\oint_s \vec{B} \cdot dA = \nabla \cdot \vec{B} = 0$$

Total magnetic flux through any closed surface is zero.

Flux of the magnetic field through an infinitesimal area element dA

Consider a bar magnet enclosed by a Gaussian surface. The magnetic field lines form closed loops, and the total flux through the surface is zero.
No matter how you position the surface, the number of field lines entering equals the number leaving.



Maxwell's Equation 2: Gauss's law of electrostatic

Differential Form

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

Electric field

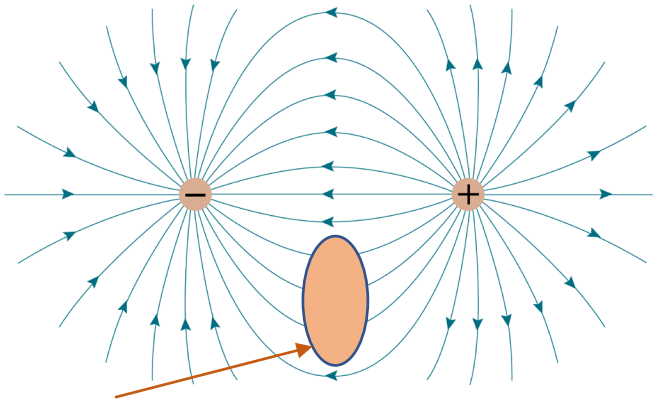
Charge density (电荷密度) $\rho = \frac{dq}{dv}$

Permittivity (介电常数，电容率，诱电率) 电介质是在电场作用下能建立极化的物质。介电常数是描述某种材料放入电容器中增加电容器存储电荷能力的物理量，是相对介电常数与真空中绝对介电常数乘积。

In general form: $\nabla \cdot \vec{D} = \rho$ 电感应强度 $\vec{D} = \varepsilon_0 \vec{E}$

Positive charges act as sources of electric field lines (field lines radiate outward).
Negative charges act as sinks of electric field lines (field lines converge inward).

If there is no net charge inside the closed surface, the total electric flux through the surface is zero. This does not mean the electric field is zero everywhere.



$\nabla \cdot \vec{E} = 0$ means no charge inside the closed surface S, 闭合曲面形成体积(气球)内无电荷

The integral of electric flux density is always equal to charge enclosed over a closed surface. It implies that electric field lines originate from positive charges and terminate on negative charges. (电场的**源**与**汇**大小正比于电荷密度)

Maxwell's Equation 2: Gauss's law of electrostatic

Integral Form

$$\oint_S E \cdot \mathrm{d}S = \frac{Q}{\varepsilon_0}$$

Total electric charge enclosed by the surface S

面积分，表示电场穿过闭合曲面S的通量

The integral form states that the electric flux through any closed surface is proportional to the net charge enclosed by the surface.

Differential Form

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

Integral Form

$$\oint_S E \cdot \mathrm{d}S = \frac{Q}{\varepsilon_0}$$

$$\int_v \nabla \cdot \vec{E}\, \mathrm{d}V = \int_v \frac{\rho}{\varepsilon_0} \mathrm{d}V = \oint_S E \cdot \mathrm{d}S = \frac{Q}{\varepsilon_0}$$

散度的体积分，即体积V中所有源点和汇点的总流量

Divergence Theorem（散度定理）：散度体积分等效于面积分。Total net flow generated inside the volume (volume integral of divergence) equals the total flow exiting or entering the surface enclosing the volume (surface integral of the field).

Other form derivation

$$\int_V (\nabla \cdot \mathbf{D}) dV = \int_V \rho_V dV$$

$$\Rightarrow \int_S \mathbf{D} \cdot \mathbf{dS} = Q_{enc}$$

电感应强度 $\vec{D} = \varepsilon_0 \vec{E}$

Maxwell’s Equation 3

Faraday’s law of electromotive force （法拉第感应定律：时变磁场产生电场）

Michael Faraday, a self-taught scientist, conducted a series of experiments (1831) that led to the discovery of electromagnetic induction.

(1)The Induction Ring: Faraday wound two coils of wire around an iron ring. When he connected one coil to a battery, creating a current, he observed a transient current in the second coil, even though it was not directly connected to the battery.

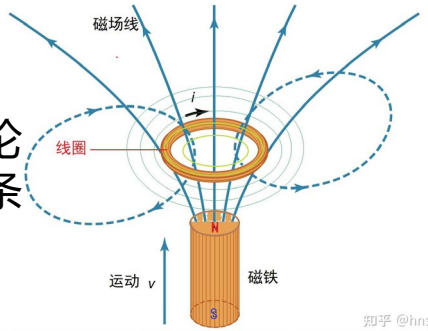
(2) Magnetic Flux and Motion: Faraday also experimented with moving magnets near coils of wire. He found that moving a magnet through a coil or moving a coil relative to a magnet induced an EMF in the coil.

Based on his experiments, Faraday formulated the principle of electromagnetic induction: "An electromotive force is induced in a conductor whenever there is a change in the magnetic flux linked with the conductor."

Lenz's Law

Heinrich Lenz later clarified the direction of the induced current, stating that the induced EMF always opposes the change in magnetic flux that caused it. This is now known as Lenz's Law , which explains the negative sign in Faraday's equation.

楞次总结了安培的电动力学与法拉第的电磁感应现象后，于1833年在圣彼得堡科学院宣读了题为“关于用电动力学方法决定感生电流方向”的论文，提出了感应电动势阻止产生电磁感应的磁铁或线圈的运动，后来这条定律被称为楞次定律，在1834年的《物理学和化学年鉴》上发表。随后德国物理学家亥姆霍兹证明楞次定律实际上是电磁现象的能量守恒定律。

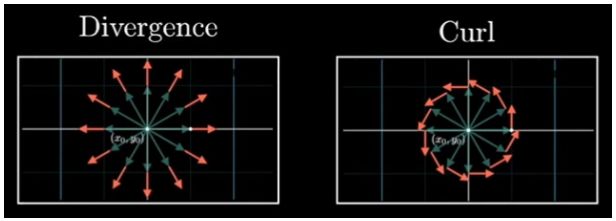


知乎 @hnsjc

Maxwell’s Equation 3

Faraday’s law of electromotive force （法拉第感应定律：时变磁场产生电场）

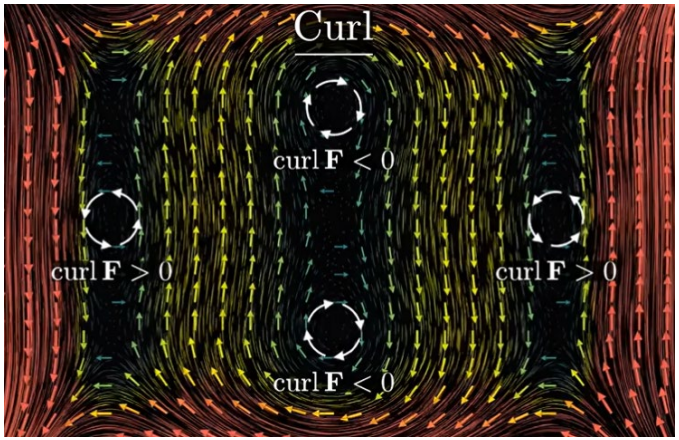
Faraday's experimental findings were later formalized mathematically by James Clerk Maxwell in the 1860s.

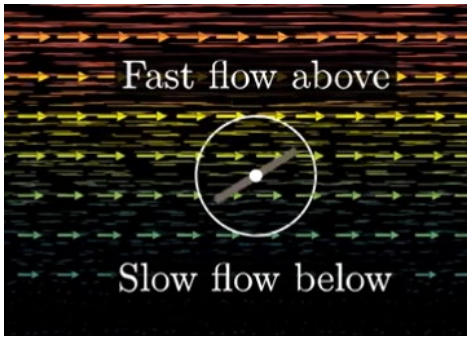


$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

叉乘将微分算子del
转换为Curl 旋度

旋度是描述向量场中是否存在“旋转”或“涡旋”的趋势。





一侧快，另一侧慢（颜色代表速度），则该点也是有旋的。

Time-varying magnetic field will always produce an electric field.
随时间变化的磁场会产生环绕的涡旋电场

Maxwell’s Equation 4

Ampere’s Circuital law （安培定律：时变电场产生磁场）

1820s，Ampère conducted experiments with two parallel current-carrying wires and observed that they exerted forces on each other. Ampère generalized his findings into a mathematical law, now known as **Ampère's Circuital Law** :

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

where: **B** is the magnetic field, *C* is a closed loop, *d***l** is an infinitesimal segment of the loop, I_{enc} is the total current enclosed by the loop, μ_0 is the permeability of free space.

The law states that the line integral of the magnetic field around a closed loop is proportional to the total current passing through the surface bounded by the loop.

The original form of Ampère's law did not account for the displacement current, which arises when electric fields change with time.

Maxwell introduced the concept of **displacement current** , which accounts for the changing electric flux through a surface:

$$I_{\text{displacement}} = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Leading to the modern (integral) form of the equation：

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(I_{\text{enc}} + \varepsilon_0 \frac{d\Phi_E}{dt} \right)$$

Maxwell’s Equation 4

Differential Form of the equation: $$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

从积分形式到微分形式的转换

（1）应用斯托克斯定理

斯托克斯定理将环路积分与面积分联系起来：

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{A}$$

其中：

- S 是闭合路径 C 所包围的任意曲面，
- $d\mathbf{A}$ 是曲面上的无穷小面积元。

将斯托克斯定理代入安培定律的积分形式：

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{A} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{A} + \mu_0 \varepsilon_0 \int_S \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{A}$$

（2）消去面积分

由于上述等式对任意曲面 S 都成立，因此被积函数必须相等：

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

Maxwell’s Equation 4

Ampere’s Circuital law （安培定律：时变电场产生磁场）

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

→ 电场随时间变化率

Permeability /ˌpɜːmiəˈbɪləti/ (磁导率): the measure of magnetization produced in a material in response to magnetic field.

Current density

Permittivity (介电常数)

电流和随时间变化的电场(移动电场)会激发出环绕的涡旋磁场

Speed of light

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$$

ε_0 permittivity of free space 8.8510^{-12} $\frac{farad}{m}$

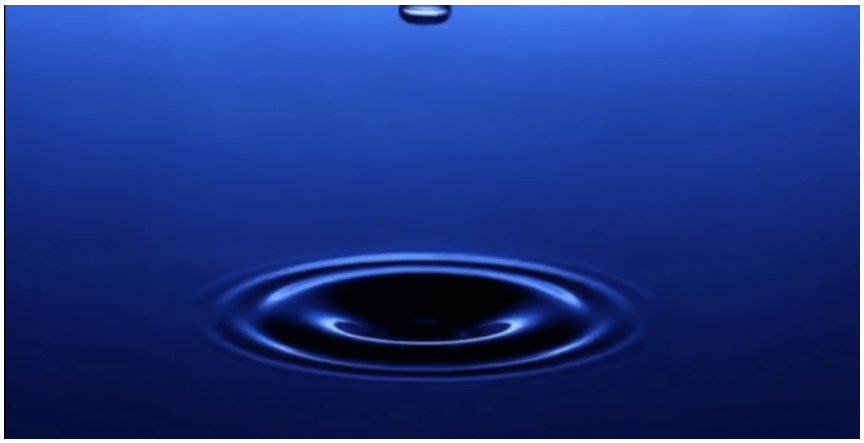
μ_0 permeability of free space $4\pi 10^{-7}$ $\frac{henry}{m}$

$$c = \sqrt{\frac{1}{(\varepsilon_0 \mu_0)}} = 2.99792458 \ x \ 10^8 \, m/s$$

μ 和 ε 分别反映了电场和磁场在空间中的“惯性”和“弹性”，它们的乘积决定了电磁波传播的时间尺度。

Permittivity ε_0 and Permeability μ_0 can be thought as the resistance of free space against the formation of electric fields and magnetic fields, respectively. Like viscosity in water.

Wave optics: What is a wave?



In the mathematical sense, a wave is any function that moves.
Define the concept of a wave: function of space (x) and time (t).

One space dimensional (only in x direction) wave equation in simple form:
Forward f(x-vt) and backward f(x+vt), where v is phase velocity of the wave.

But…why do I always see wave equations like these

What are they???

$$\frac{\partial^2 E}{\partial x^2} - \mu\varepsilon \frac{\partial^2 E}{\partial t^2} = 0$$

$$E(x,t) = A\cos[(kx - \omega t) - \delta]$$

$$E(x,t) = E_0 \mathrm{e}^{j(kx-\omega t)}$$

Partial derivatives

For example, if $f(x,t) = e^{ax} + e^{bt}$, then: $\frac{\partial f}{\partial x} = ae^{ax}$ and $\frac{\partial f}{\partial t} = be^{bt}$

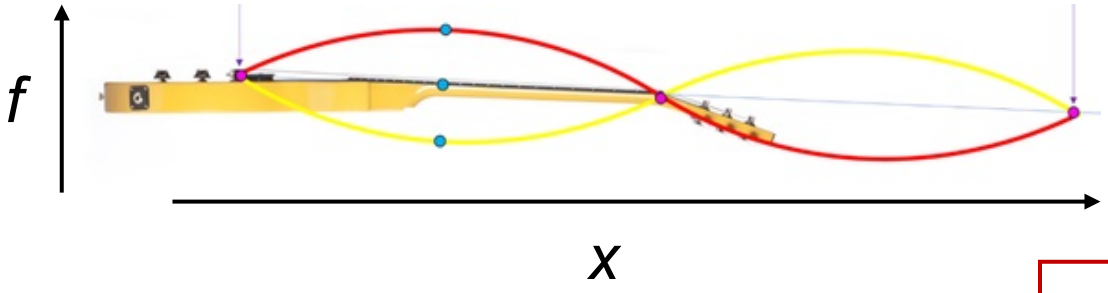
$\frac{\partial f}{\partial t}$ This is called a partial derivative (偏导)

$\frac{\partial^2 E}{\partial x^2}$ Second partial derivatives (二阶偏导数)

This d-like character has been around for two hundred years, but, strangely, it has no actual name, so we just say, “the partial derivative of f with respect to x or t”.

That is a mouthful, so we can just call it “d” because it’ll usually be clear from the slide what it means.

General wave equation



A guitar string wants to be straight; the restoring force is proportional to how much it's stretched or curvature:

$$F = \gamma \frac{\partial^2 f}{\partial x^2}$$

where γ is constant 张力, *f(x,t)* is the displacement (弦中横向位移) from straight at the point *x* at time *t*. f

But Newton's 2nd Law ($\boldsymbol{F = ma}$) says that this equals:

$$F = m \frac{\partial^2 f}{\partial t^2}$$

Setting them equal:

$$\frac{\partial^2 f}{\partial x^2} - \frac{1}{\mathrm{v}^2}\frac{\partial^2 f}{\partial t^2} = 0$$

where: $\mathrm{v} = \sqrt{\gamma / m}$

Waves are a solution to this equation. And v is the wave's velocity, also called its phase velocity.

Proof that f (x ± vt) solves the wave equation

Write $f(x \pm vt)$ as $f(u)$, where $u = x \pm vt$. So $\frac{\partial u}{\partial x} = 1$ and $\frac{\partial u}{\partial t} = \pm v$

Use the chain rule:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} = \frac{\partial f}{\partial u} \qquad \frac{\partial f}{\partial t} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial t} = \pm v\frac{\partial f}{\partial u}$$

Computing the second derivatives adds another factor of $\partial u/\partial x$ or $\partial u/\partial t$:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial u^2} \qquad \frac{\partial^2 f}{\partial t^2} = v^2\frac{\partial^2 f}{\partial u^2}$$

Substituting into the wave equation:

$$\frac{\partial^2 f}{\partial x^2} - \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} = \frac{\partial^2 f}{\partial u^2} - \frac{1}{v^2}\left\{v^2\frac{\partial^2 f}{\partial u^2}\right\} = 0$$

Electromagnetic wave equation

$$\frac{\partial^2 f}{\partial x^2} - \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} = 0$$

f(x,t) is the displacement

v is the wave's velocity

$$\frac{\partial^2 E}{\partial x^2} - \mu\varepsilon\frac{\partial^2 E}{\partial t^2} = 0$$

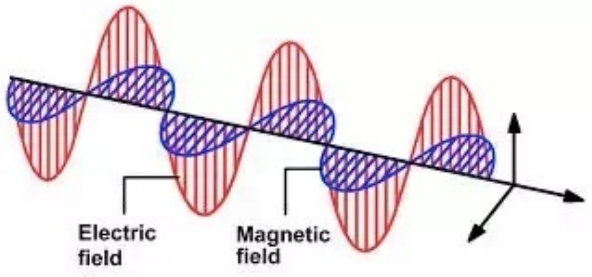
E(x,t) is electric field,

μ is permeability 磁导率,

ε is permittivity 介电常数

$c = \frac{1}{\sqrt{\mu\varepsilon}}$ Light velocity

时变电场和磁场相互激发，有一个π/2的相位差。电场激发的磁场往前移了λ/4个波长的位置，磁场激发电场也是一样，相互激发往前传播。



We derive the electromagnetic wave equation from Maxwell's equations.

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Electromagnetic wave equation

Taking the curl of both sides: $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\frac{\partial \vec{B}}{\partial t}\right)$ **1**

The right-hand side is simplified by noting that:

$\nabla \times$ is for spatial operation, do not related to $\frac{\partial}{\partial t}$

$$\nabla \times \left(-\frac{\partial \vec{B}}{\partial t}\right) = -\frac{\partial}{\partial t}(\nabla \times \vec{B})$$ **2**

Recalling Maxwell's Equation 4 (Ampère's Law)

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Substituting $\nabla \times \vec{B}$ in eq.2 :

$$\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

The left hand side is simplified by the following vector identity （向量恒等式）:

$$\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$$

For a constant field or no charges $\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} = 0$ Hence $\nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$

$$\nabla^2 \vec{E} = \frac{\partial^2 \vec{E}}{\partial x^2}$$

Laplace operator, Laplacian
拉普拉斯算子：函数的梯度的散度给出的微分算子，通常写成 Δ或 ∇^2，二阶偏导数

Therefore $\boxed{\frac{\partial^2 \vec{E}}{\partial x^2} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}}$

Harmonic wave solutions to the wave equation

Harmonic solutions 是波动方程的特定解，描述了在空间和时间上呈正弦或余弦形式的周期性波。这些解通常被称为简谐波解，因为它们具有简单的正弦或余弦形式，并且是波动方程中最基本、最重要的解之一。

Harmonic waves are sinusoidal solutions to the wave equation. They are periodic in both space and time and can be written in the general form:

$$E(x,t) = A\,cos[(kx - \omega t) - \delta]$$

or equivalently: $E(x,t) = E_0 e^{j(kx-\omega t)}$

- A is the amplitude of the wave,
- k is the **wave number** , related to the wavelength λ by $k=\lambda 2\pi$,
- ω is the **angular frequency** , related to the frequency f by $\omega=2\pi f$,
- δ is the phase constant,
- $v=k\omega$ is the wave speed.

Harmonic wave solutions to the wave equation

$$\frac{\partial^2 E}{\partial x^2} - \mu\varepsilon \frac{\partial^2 E}{\partial t^2} = 0$$

The most general solution has **two** unknown constants, which cannot be determined without some additional information about the problem.
We might expect that oscillatory solutions (**sines** and **cosines**) will be very relevant for light waves. (二阶线性齐次微分方程通解：)

$$E(x,t) = B\cos[k(x \pm ct)] + C\sin[k(x \pm ct)]$$

Making the substitution $kc = \omega$

$$E(x,t) = B\cos(kx \pm \omega t) + C\sin(kx \pm \omega t)$$

A simpler equation for a harmonic wave

二阶线性齐次微分方程（波函数）通解：

$$E(x,t) = B\cos(kx \pm \omega t) + C\sin(kx \pm \omega t)$$
$$B = A\cos(\theta) \qquad C = A\sin(\theta)$$

$$E(x,t) = A\cos(kx \pm \omega t)\cos(\theta) + A\sin(kx \pm \omega t)\sin(\theta)$$

根据三角恒等式化解上述公式Trigonometric identity:

$$\cos(\alpha - \beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)$$

where $\alpha = kx \pm \omega t$ and $\beta = \theta$.

Common way of writing the solution to the wave equation:

$$E(x,t) = A\cos[\underbrace{(kx \pm \omega t)}_{\alpha} - \underbrace{\theta}_{\beta}]$$

A = **Amplitude**, θ = **Absolute phase** (or “initial” phase t = 0)

Forward and backward-propagating wave

$$E(x,t) = A\cos[(kx \pm \omega t) - \theta]$$

where ω is a positive constant and t is time, then the displacement of this function will increase with time when $kx-\omega t$

So $kx - \omega t$ represents a rightward, or forward propagating wave.

Similarly,$kx + \omega t$ represents a leftward, or backward propagating wave.

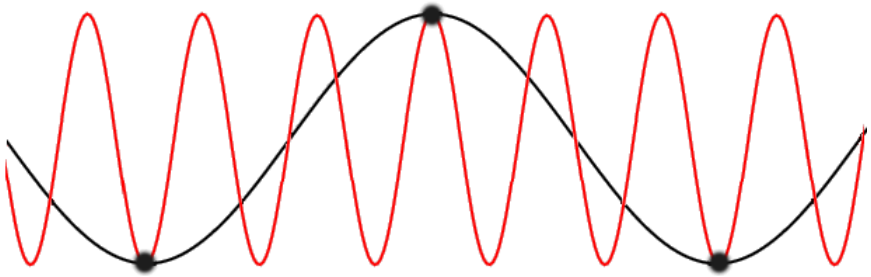
For simplicity, use only the forward-propagating wave:

$$E(x,t) = A\cos[(kx - \omega t) - \theta]$$

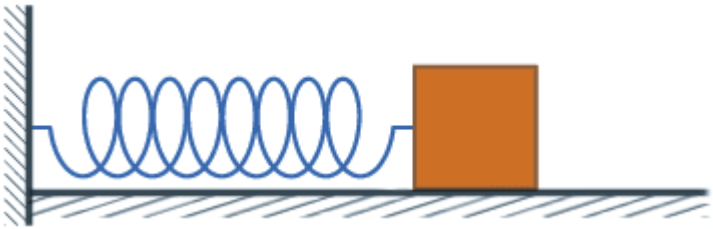
Waves and the Wave Equation

Wave is any function that moves

$$E(x,t) = A\cos[(kx - \omega t) - \theta]$$



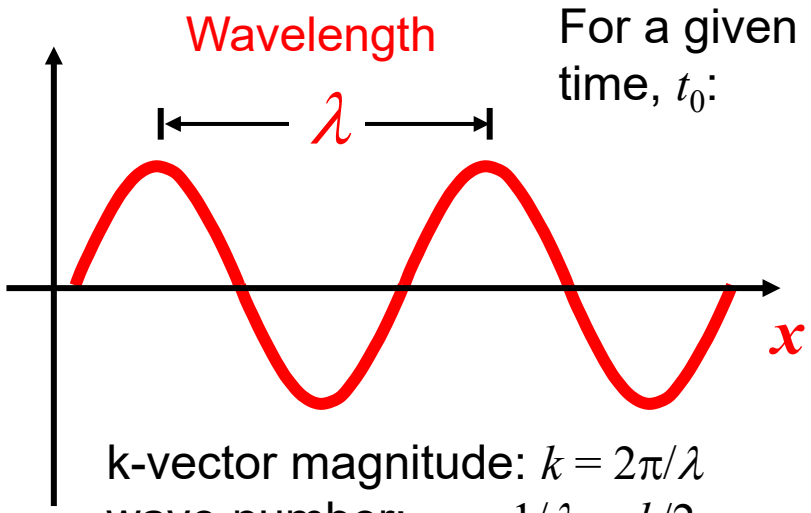
Simple harmonic motion (一维简谐振动) 常用于描述质点的简单谐振动，而波动方程更广泛地描述了波的传播和干扰现象。



$$x(t) = A\cos(\omega t + \varphi)$$

Wavelength in spatial and temporal quantities

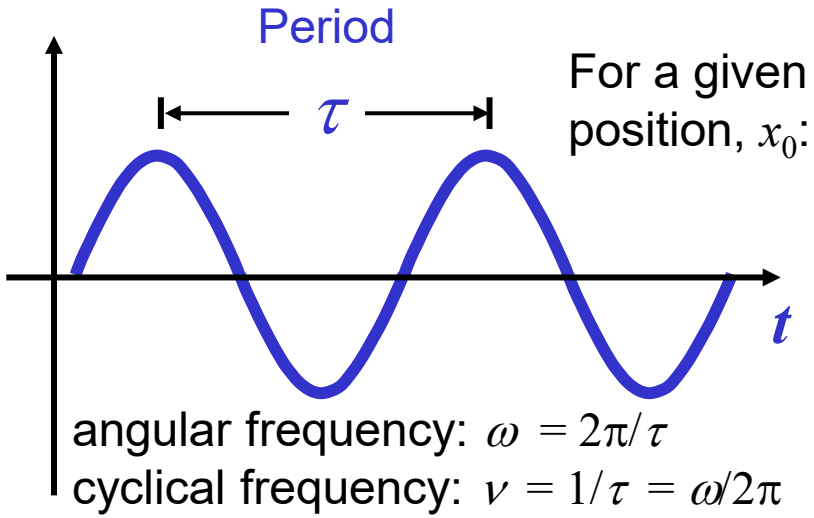
Spatial quantities:



k-vector magnitude: $k = 2\pi/\lambda$

wave number: $\kappa = 1/\lambda = k/2\pi$

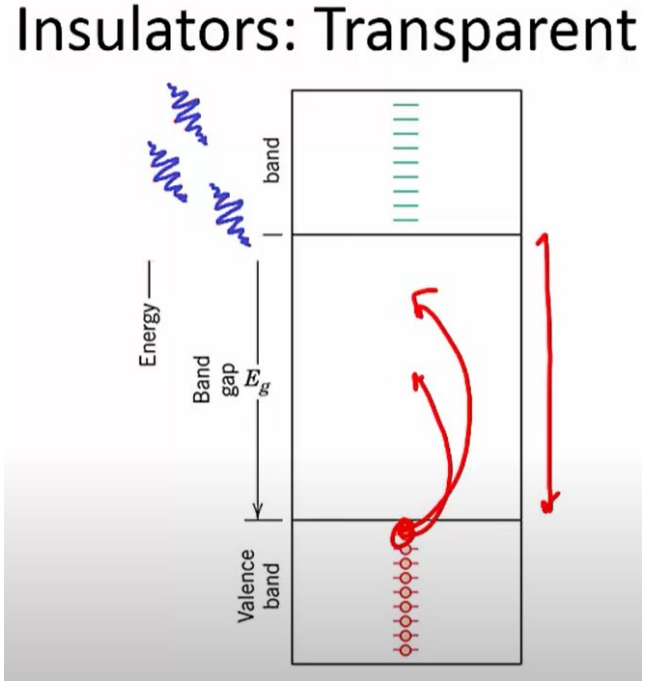
Temporal quantities:



angular frequency: $\omega = 2\pi/\tau$

cyclical frequency: $\nu = 1/\tau = \omega/2\pi$

Light absorption in matter: why light slow down?





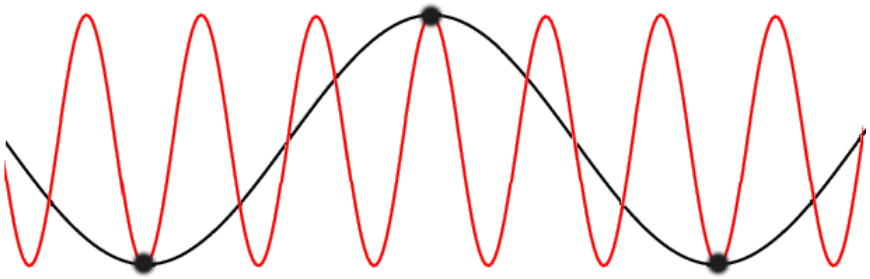
Insulators tend to have a very wide band gap, larger than the energy of visible photons. The light waves are absorbed and then re-emitted by the insulators.

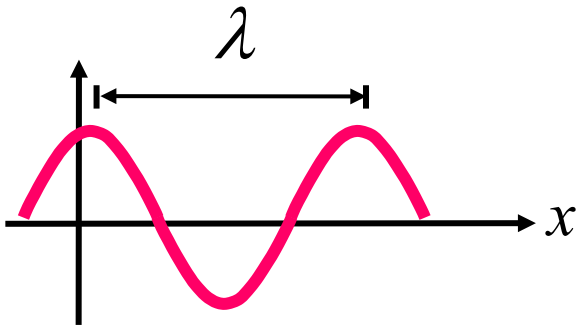
当电磁波进入介质时，电场会使介质中的电子发生振动（极化）。这些振动电子会重新辐射出次级波，这些次级波与入射波叠加，改变了波的整体传播特性。由于电子的振动不是瞬时的，而是有一定的滞后性，这导致了波的相位延迟。

色散:不同频率的波在介质中受到的响应不同，导致它们的相速度不同。

Wave Velocity

Wave is any function that moves





Note: if you take a snapshot at any instant of time, the magnitude of E oscillates as a function of x.

Wave moves one wavelength, λ, in one period, τ.

And: if you sit at any location x, the magnitude of E oscillates as a function of time t.

In terms of the k-vector, $k = 2\pi/\lambda$,表示波的空间和方向，单位长度内的波数
波长越长,波矢越小。
and the angular frequency, 角频率 $\omega = 2\pi f$, 每个点的振动频率。
therefore light velocity, $c = \lambda f = \omega/k$, connect by frequency.

Wave Velocity

在介质中，波的电场或磁场会引起介质内部的极化或磁化响应。这种响应不是瞬时的，而是需要一定的时间（称为弛豫时间 ）。因此，波的相位在介质中会“滞后”于它在真空中的相位。相位延迟意味着波的传播在介质中被“拖慢”了。

根据波速公式 $c = \omega/k$ ，如果相位延迟导致波数 k 增大（波长缩短），而角频率 ω 不变，则波速 c 会减小。

介质对波的影响通常用折射率 n 来描述。折射率定义为： $n = c/v$

c 是光在真空中的速度，v 是波在介质中的相速度。
当波进入介质时，折射率 n>1，说明波速 v<c。

Waves using complex numbers

We have known the Harmonic Wave Solutions to the Wave Equation:

$$E(x,t) = A\cos[(kx - \omega t) - \theta]$$

What is this? Why the wave equation using complex numbers:

$$E(x,t) = E_0 \mathrm{e}^{j(kx-\omega t)}$$

We need to know **Euler's formula**

Euler's formula

Euler's formula, named after Leonhard Euler, is a mathematical formula in complex analysis that establishes the fundamental relationship between the trigonometric functions and the complex exponential function.

$$e^{ix} = cosx + isinx$$

When x = π, Euler's formula may be rewritten as $e^{i\pi}$ + 1 = 0 or $e^{i\pi}$ = -1, which is known as Euler's identity(欧拉恒等式).

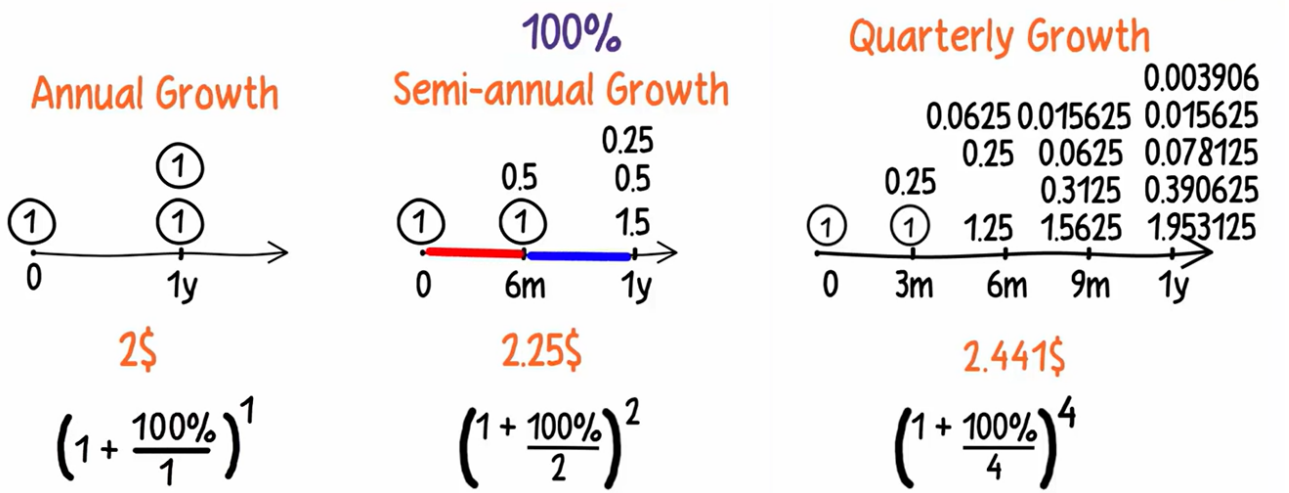
The number e is sometimes called Euler's number. It is of great importance in mathematics, alongside 0, 1, π, and i. All five appear in one formulation of Euler's identity and play important and recurring roles across mathematics.

The number e is a mathematical constant approximately equal to 2.71828. It is the base of natural logarithms. It is the limit of (1 + 1/n)n as n approaches infinity. The number e

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n} = 2.71828\ldots$$

Background of Euler's formula: What is "e"?

"e" constant was discovered by the Swiss mathematician Jacob Bernoulli while studying compound interest in 1683.



100% Interest (per year): $P(1+1) = 2P$

50% Interest (per $\frac{1}{2}$ year): $P\left(1+\frac{1}{2}\right)^2 = 2.25P$

33% Interest (per $\frac{1}{3}$ year): $P\left(1+\frac{1}{3}\right)^3 = 2.37P$

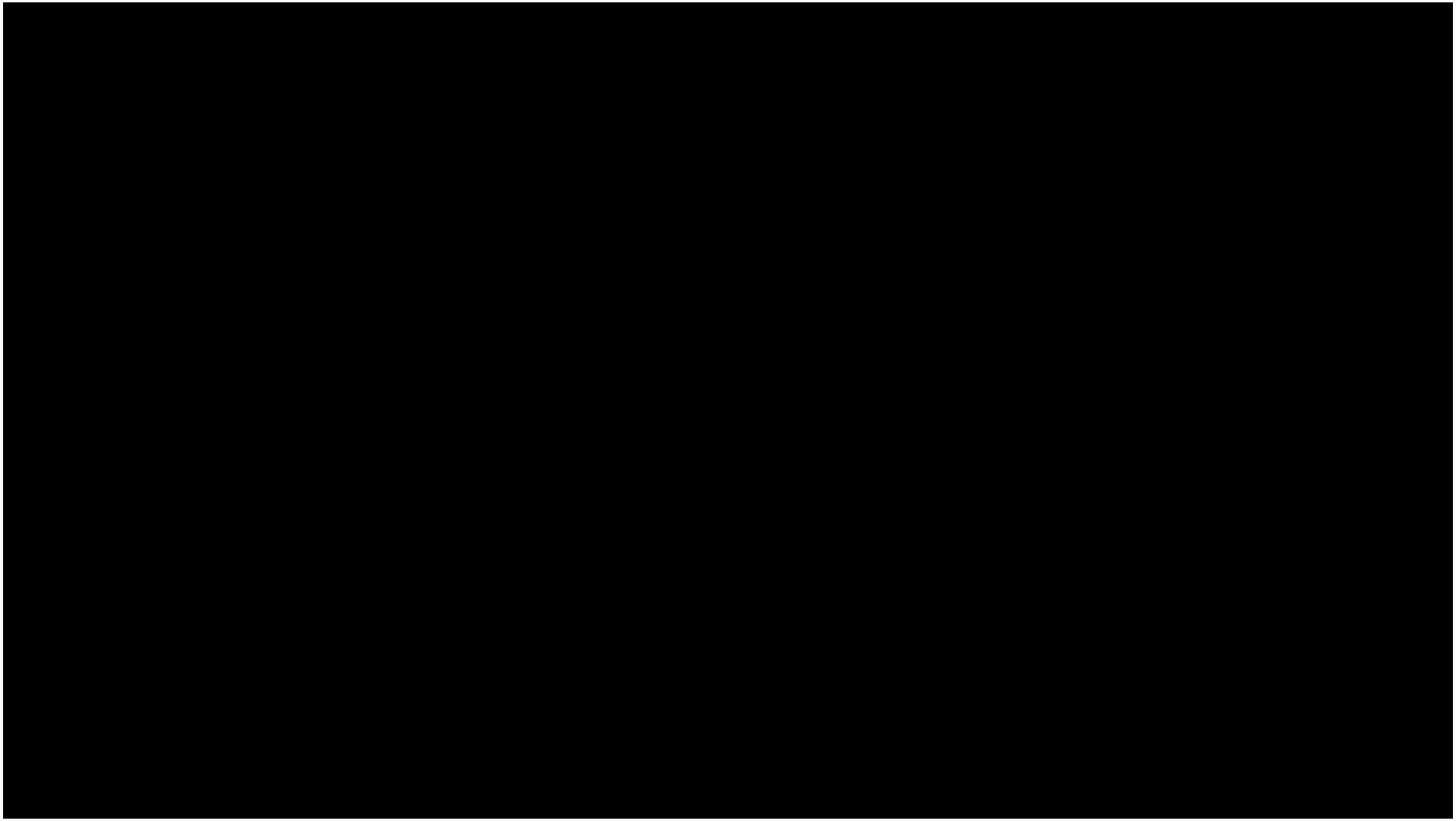
$\vdots$

$\frac{100}{n}\%$ Interest (per $\frac{1}{n}$ year): $P\left(1+\frac{1}{n}\right)^n$

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = 2.71828\ldots$$

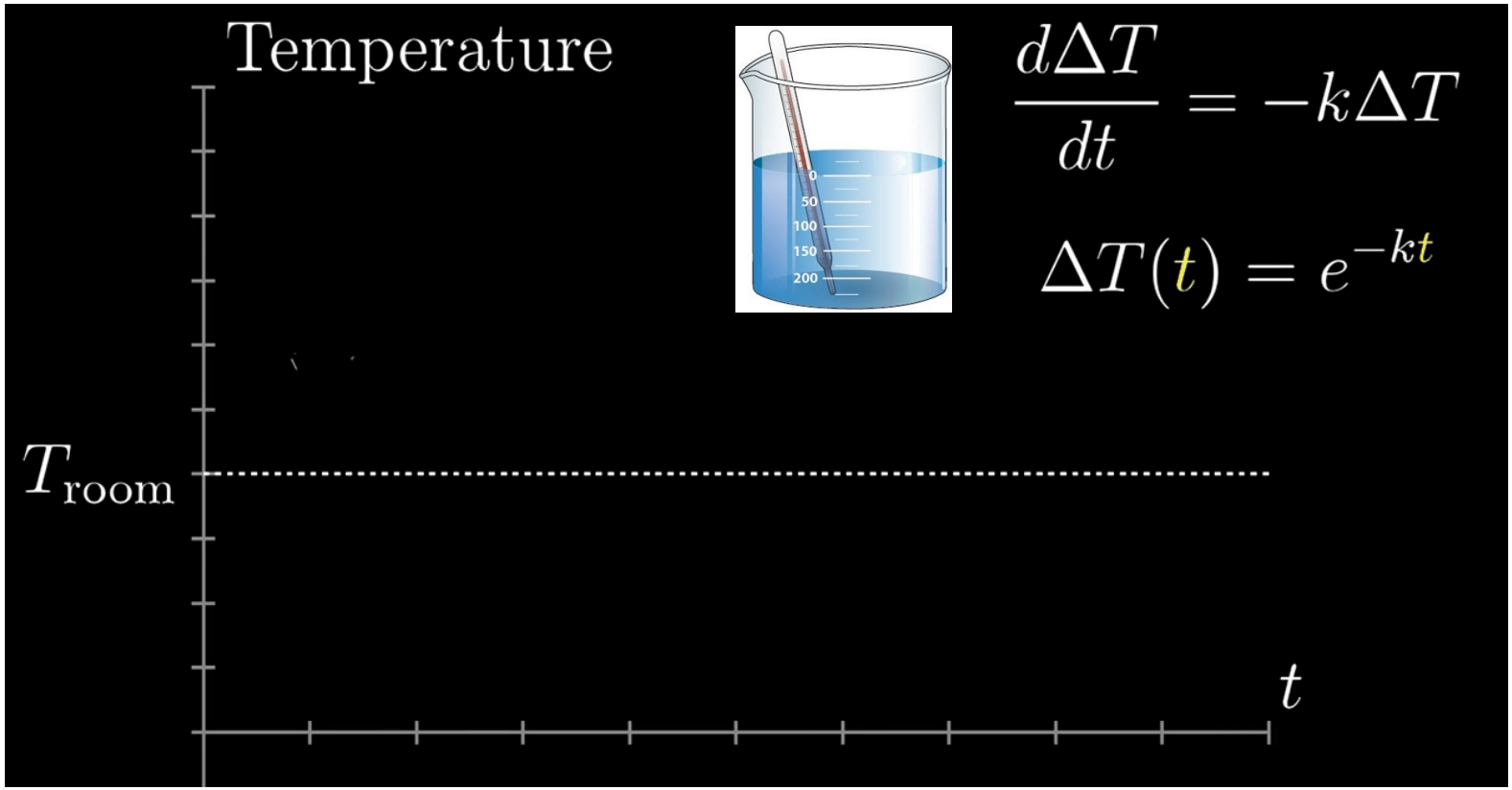
Background of Euler's formula: What is "e"?

What do these equations mean? $\frac{d}{dx}e^{x} = e^{x}$ $\frac{d}{dx}e^{ax} = ae^{x}$



Why “e” is important?

$$\frac{d}{dx}\mathrm{e}^{x} = \mathrm{e}^{x} \qquad \frac{d}{dx}\mathrm{e}^{ax} = \mathrm{a}\mathrm{e}^{x}$$



Proof of Euler's Formula

We have learned what is "e",
but why :

$$e^{ix}$$

equals to

$$cosx + isinx$$

Let's proof the Euler's Formula: $e^{ix} = cosx + isinx$

Taylor Series and Approximations

First, we should know Taylor Series:

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ldots$$

Example:

$$\exp(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$$

because all derivatives are also $\exp(x)$ and $\exp(0) = 1$

If x is close to zero (i.e., << 1), then we can often approximate the function with the first one or two nonzero terms of its Taylor series:

$$\exp(x) \approx 1 + x$$

Other functions it'll be convenient to approximate in this manner:

$$\exp(-x^2) \approx 1 - x^2$$

$$\frac{1}{1+x} \approx 1 - x$$

$$\frac{1}{\sqrt{1-x}} \approx 1 + \tfrac{1}{2}x$$

$$\frac{1}{\sqrt{1-x^2}} \approx 1 + \tfrac{1}{2}x^2$$

https://www.youtube.com/watch?v=3d6DsjIBzJ4 第13分钟，为什么麦克劳林展开x=0可行

Proof of Euler's Formula: $\exp(i\varphi) = \cos(\varphi) + i\sin(\varphi)$

Use Taylor Series: $$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ldots$$

$$\exp(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} + \ldots$$

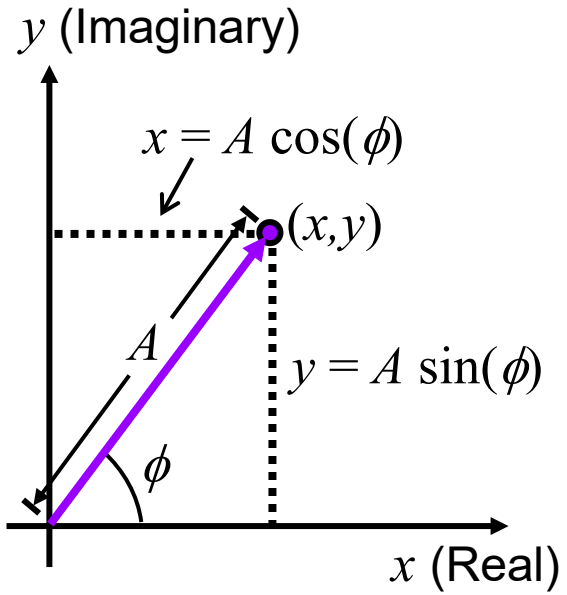
$$\sin(x) = \frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} + \ldots$$

If we substitute $x = i\varphi$ into $\exp(x)$, then:

$$\begin{aligned}\exp(i\varphi) &= 1 + \frac{i\varphi}{1!} - \frac{\varphi^2}{2!} - \frac{i\varphi^3}{3!} + \frac{\varphi^4}{4!} + \ldots \\ &= \left[1 - \frac{\varphi^2}{2!} + \frac{\varphi^4}{4!} + \ldots\right] + i\left[\frac{\varphi}{1!} - \frac{\varphi^3}{3!} + \ldots\right] \\ &= \cos(\varphi) + i\sin(\varphi)\end{aligned}$$

Complex-number

Consider a point, **(x,y)** on a 2D Cartesian grid.



Let the x-coordinate be the real part and the y-coordinate the imaginary part of a **complex number**:

$$z \equiv x + i\,y$$

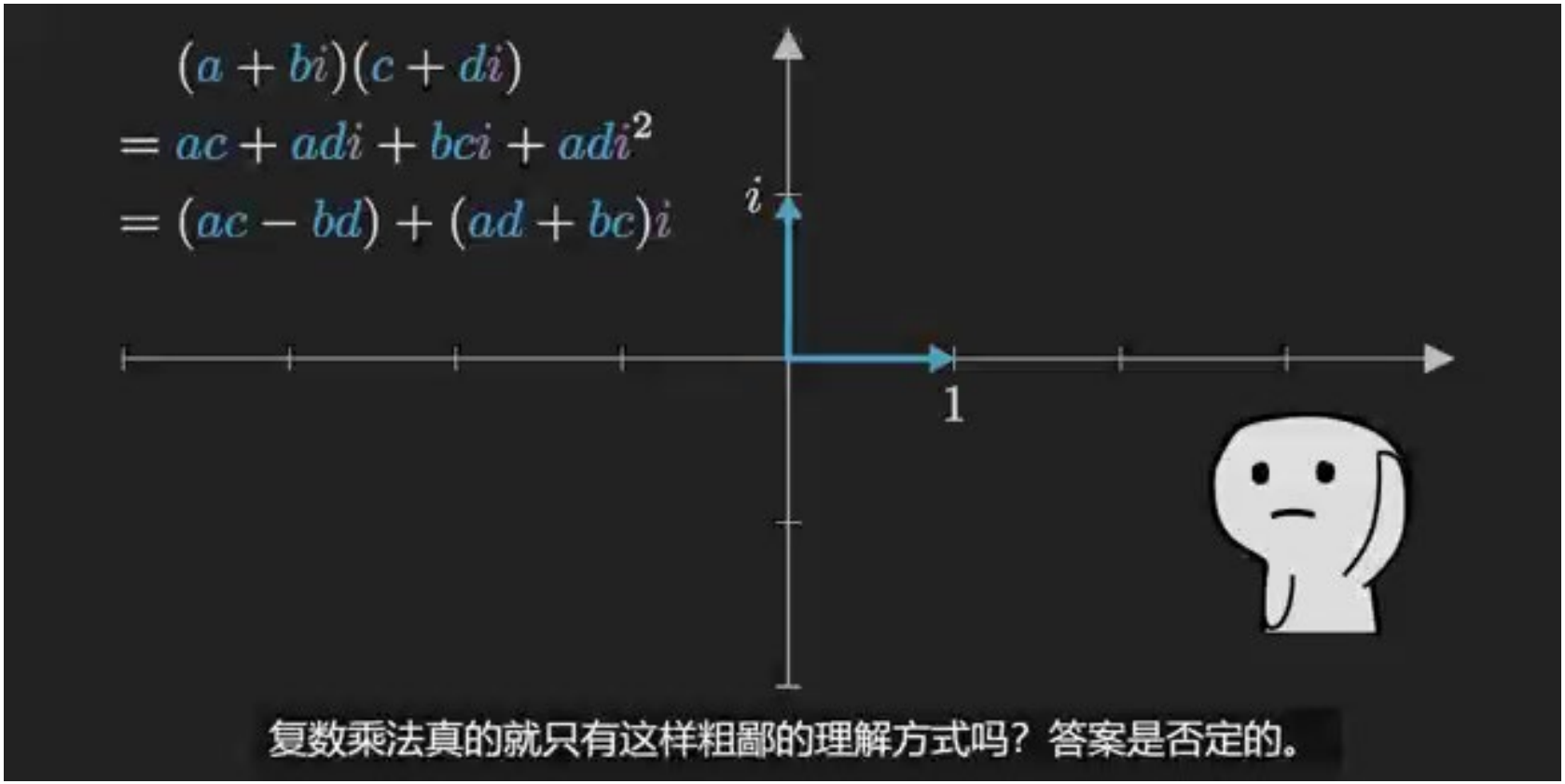
$$= A\cos(\phi) + i\,A\sin(\phi)$$

where $i = \sqrt{-1}$

Euler's Formula

$$e^{ix} = cosx + isinx$$

Euler's formula

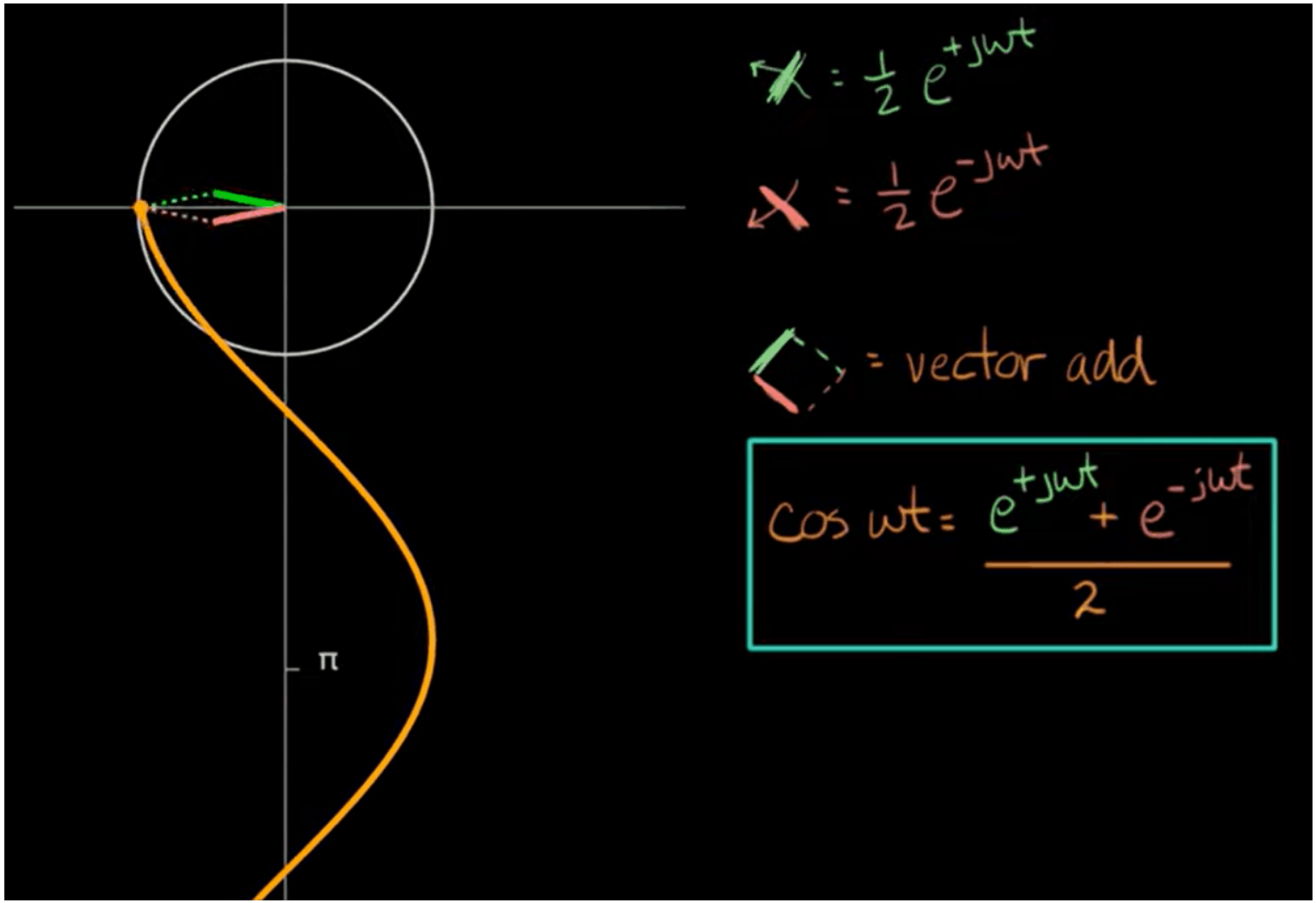


第10分钟开始

Euler's formula

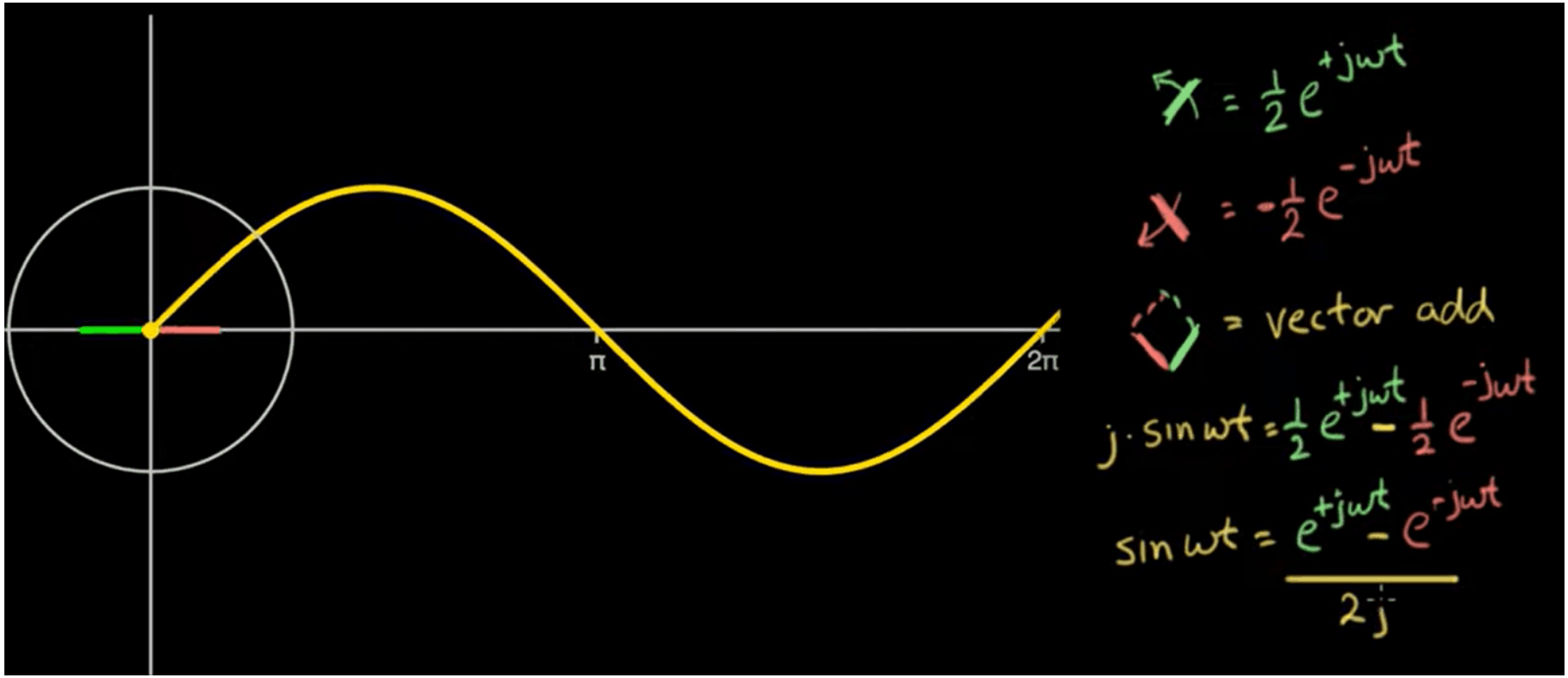
$$e^{j\omega t} = cos\omega t + jsin\omega t$$

In physics: use $i = \sqrt{-1}$, and j = current density
In engineering: use $j = \sqrt{-1}$, and i = current



Euler's formula

$$e^{j\omega t} = cos\omega t + jsin\omega t$$



***j* means rotate 90 degrees**

Waves using complex numbers

We have known the Harmonic Wave Solutions to the Wave Equation:

$$E(x,t) = A\cos[(kx - \omega t) - \theta]$$

What is this? Why the wave equation using complex numbers:

$$E(x,t) = E_0 e^{j(kx-\omega t)}$$

We need to know **Euler's formula**

Waves and the Wave Equation

$$E(x,t) = A\cos[(kx-\omega t)-\theta]$$

$$\cos(kx-wt-\theta) = \frac{e^{i(kx-\omega t-\theta)} + e^{-i(kx-wt-\theta)}}{2}$$

$$\sin(kx-wt-\theta) = \frac{e^{i(kx-\omega t-\theta)} - e^{-i(kx-wt-\theta)}}{2i}$$

$$e^{j\varphi} = cos\varphi + isin\varphi$$

在物理中，可观测的量通常是波函数的实部或模平方，而不是复数本身。这是因为物理量（如位移、电场强度、概率密度等）必须是实数。因此，在实际应用中，我们通常只取波函数的实部：

$$E(x,t) = \mathrm{Re}\{Ae^{i(kx-\omega t-\theta)}\}$$

or $$E(x,t) = \frac{1}{2}Ae^{i(kx-\omega t-\theta)} + c.c.$$

where "$+ c.c.$" means "plus the complex conjugate of everything before the plus sign."

We often write these expressions without the ½, Re, or $+ c.c.$

$E_0 = Ae^{-i\theta}$ Does not depend on x or t

$$E(x,t) = E_0 e^{i(kx-\omega t)}$$

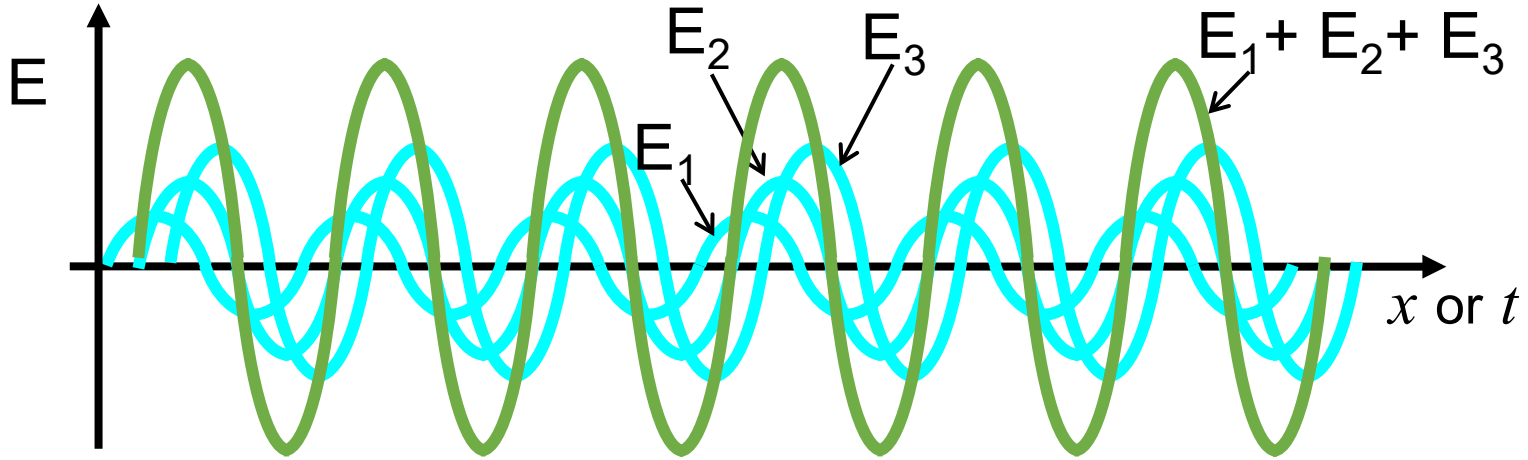
Complex numbers greatly simplify waves!

Adding waves of the same frequency, but different amplitude and/or absolute phase, yields a wave of the same frequency.

This addition is hard with trig functions, but easy with complex exps:

$$E_{tot}(x,t) = E_{01}\exp i(kx-\omega t) + E_{02}\exp i(kx-\omega t) + E_{03}\exp i(kx-\omega t)$$
$$= (E_{01}+E_{02}+E_{03})\exp i(kx-\omega t)$$

where all amplitudes and absolute phases are lumped into E_{01}, E_{02}, and E_{03}.



But this only works for *adding* waves. Don't try to *multiply* them.

Complex refractive index and light loss in metal

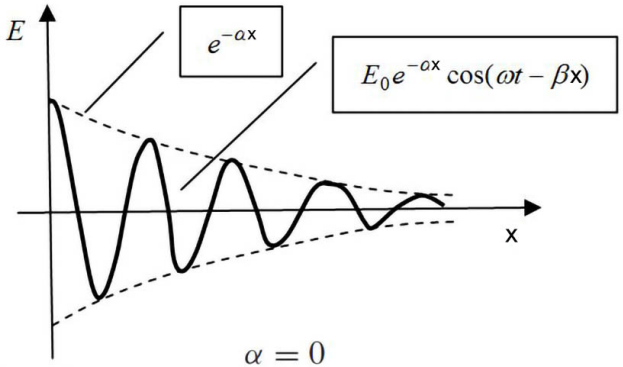
$$E(x,t) = E_0 e^{j(kx-\omega t)}$$

$k = \frac{2\pi}{\lambda}$ $k = \frac{2\pi \boldsymbol{n}}{\lambda}$ as propagation in a medium with refractive index *n*

$$n = n_0 + j\kappa$$

$$E(x,t) = E_0 e^{j[k(n_0+j\kappa)x-\omega t]} \qquad E(x,t) = E_0 e^{jkn_0x} e^{jkj\kappa x} e^{-j\omega t}$$

$$E(x,t) = E_0 e^{jkj\kappa x} e^{jkn_0x} e^{-j\omega t} \qquad e^{jkj\kappa x} = e^{-k\kappa x} \rightarrow \text{loss}$$



For the loss-free case, $\alpha = 0$

$$\beta = \omega\sqrt{\mu\varepsilon}$$

$$v = \frac{\Delta x}{\Delta t} = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\varepsilon}}$$

Complex refractive index and light loss in metal

$$E(x,t) = E_0 e^{j(kx-\omega t)}$$

$k = \frac{2\pi}{\lambda}$ $k = \frac{2\pi \boldsymbol{n}}{\lambda}$ as propagation in a medium with refractive index *n*

$n = n_0 + j\kappa$ Refractive index database

$E(x,t) = E_0 e^{j[k(n_0+jk)x-\omega t]}$ $E(x,t) = E_0 e^{jkn_0x} e^{jkj\kappa x} e^{-j\omega t}$

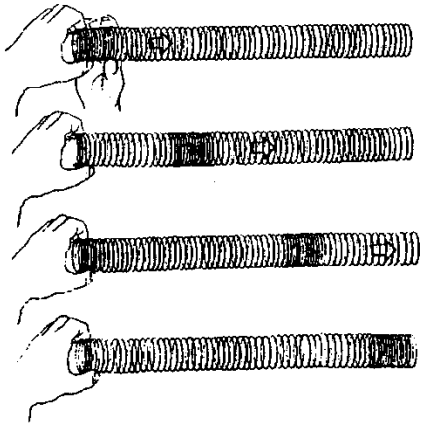
$E(x,t) = E_0 e^{jkj\kappa x} e^{jkn_0x} e^{-j\omega t}$ $e^{jkj\kappa x} = e^{-k\kappa x} \rightarrow$ loss

The complex index of refraction comes because metals have free charge carriers (actually electrons) that can oscillate in response to an incoming electromagnetic field (i.e. when you shine light on it). Those electrons move in response to the field, and re-radiate the light back - that's why metals are shiny. However, the electrons have inertia - they cannot respond instantly to the incoming field, so there is a time-lag between the incoming and outgoing electromagnetic oscillations. This time-lag (or phase difference for the technically minded) is encoded in the complex refractive index.

Longitudinal vs Transverse waves: Light polarization

The direction of a wave's variations is called its **polarization**.

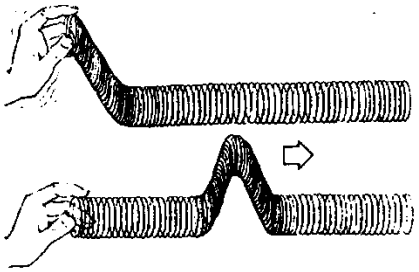
Longitudinal:



Motion is along the direction of propagation—**longitudinal polarization.**

Light will not have a longitudinal polarization.

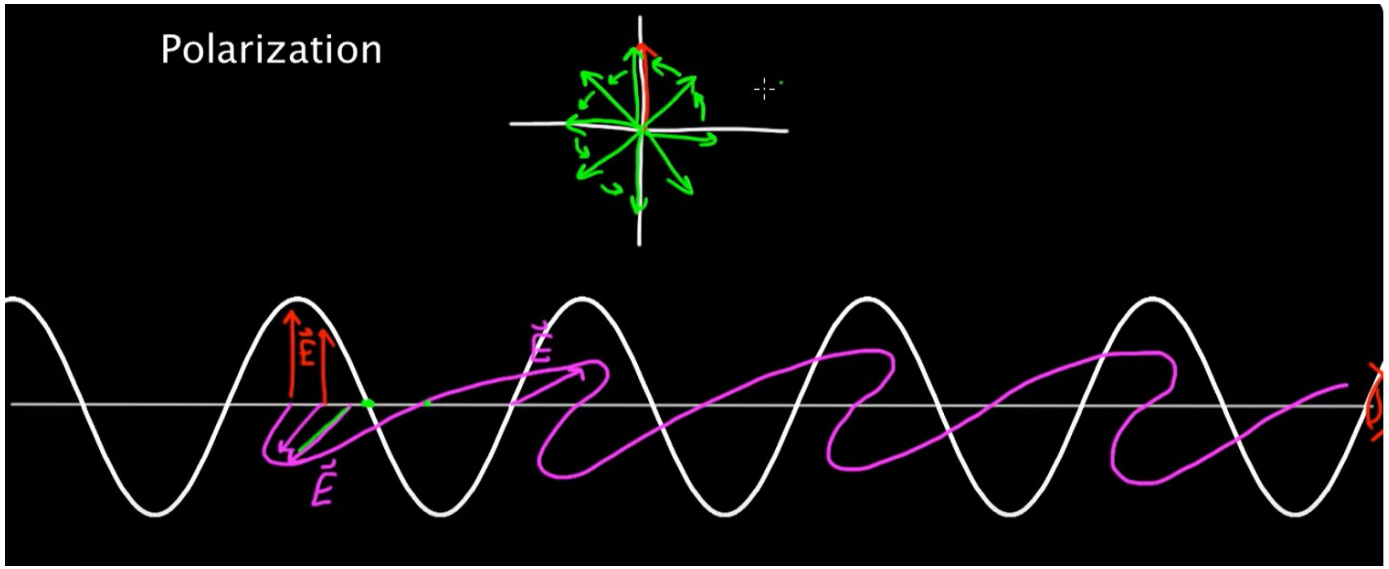
Transverse:



Motion is transverse to the direction of propagation—**transverse polarization.**

Light will always have transverse polarization.

From linear to circular polarization

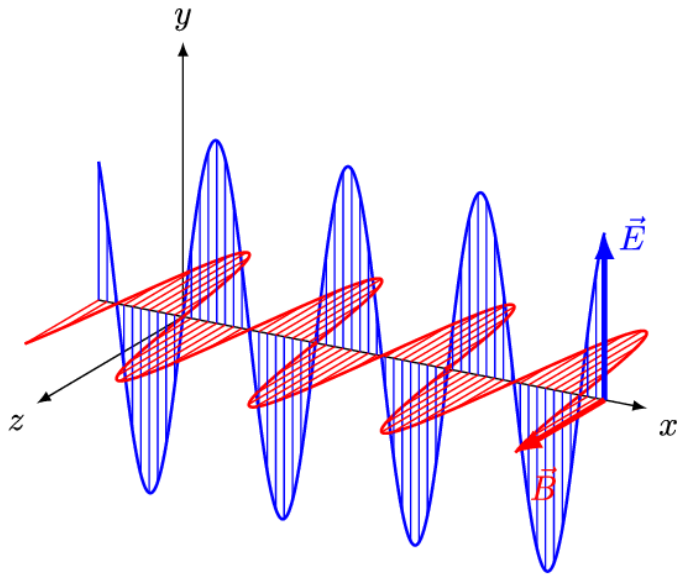


Why Light Waves are Transverse?

Suppose that an electromagnetic wave propagates in the z-direction.

Then it's of the form f(z-vt) and hence a function of z and t (and not x or y), so all x- and y-derivatives are zero.

Specifically: $\frac{\partial \mathrm{E}_x}{\partial x} = \frac{\partial \mathrm{E}_y}{\partial y} = 0$ $\qquad$ $\frac{\partial \mathrm{B}_x}{\partial x} = \frac{\partial \mathrm{B}_y}{\partial y} = 0$



Why it is zero?
E oscillates over time but the distribution of E along the x is uniform
只是在时间上变换，空间上电场磁场是均匀分布的，所以偏导是0

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Now, in a charge-free medium, $\vec{\nabla} \cdot \vec{\mathrm{E}} = 0$ $\vec{\nabla} \cdot \vec{\mathrm{B}} = 0$

that is: $\frac{\partial \mathrm{E}_x}{\partial x} + \frac{\partial \mathrm{E}_y}{\partial y} + \frac{\partial \mathrm{E}_z}{\partial z} = 0$ $\frac{\partial \mathrm{B}_x}{\partial x} + \frac{\partial \mathrm{B}_y}{\partial y} + \frac{\partial \mathrm{B}_z}{\partial z} = 0$

So: $\frac{\partial \mathrm{E}_z}{\partial z} = 0$ and $\frac{\partial \mathrm{B}_z}{\partial z} = 0$

So the longitudinal fields are **constant**—and so not waves.

The Magnetic-Field Direction in a Light Wave

Suppose a wave propagates in the z-direction and has its electric field along the x-direction.

$$\Rightarrow \quad E_y = E_z = 0 \quad \text{and} \quad E_x = E_x(z,t)$$

What's the direction of the magnetic field?

Use Faraday's Law:

$$-\frac{\partial \vec{\mathrm{B}}}{\partial t} = \vec{\nabla} \times \vec{\mathrm{E}} = \left(\frac{\partial \mathrm{E}_z}{\partial y} - \frac{\partial \mathrm{E}_y}{\partial z}, \frac{\partial \mathrm{E}_x}{\partial z} - \frac{\partial \mathrm{E}_z}{\partial x}, \frac{\partial \mathrm{E}_y}{\partial x} - \frac{\partial \mathrm{E}_x}{\partial y} \right)$$

So: $$-\frac{\partial \vec{\mathrm{B}}}{\partial t} = \left(0, \frac{\partial \mathrm{E}_x}{\partial z}, 0 \right)$$

B_x and B_z are at most constants—not waves.

and: $$-\frac{\partial \mathrm{B}_y}{\partial t} = \frac{\partial \mathrm{E}_x}{\partial z}$$

The magnetic field points in the y-direction, perpendicular to the electric field (x) and the direction of propagation (z).

Why we ignore the magnetic field?

The force on a charge, q, is

$$\vec{F} = q\vec{\mathrm{E}} + q\vec{\mathrm{v}} \times \vec{\mathrm{B}}$$

($\vec{F}_{electrical}$ = $q\vec{\mathrm{E}}$, $\vec{F}_{magnetic}$ = $q\vec{\mathrm{v}} \times \vec{\mathrm{B}}$)

where $\vec{\mathrm{v}}$ is the charge velocity

$$\left|\vec{\mathrm{v}} \times \vec{\mathrm{B}}\right| = \mathrm{vB}|\sin\theta| \leq \mathrm{vB}$$

where θ is the angle between $\vec{\mathrm{v}}$ and $\vec{\mathrm{B}}$.

Taking the ratio of the magnitudes of the two forces:

$$\frac{F_{magnetic}}{F_{electrical}} \leq \frac{q\mathrm{vB}}{q\mathrm{E}}$$

Wave Equation $E(x,t) = E_0 \mathrm{e}^{j(kx-\omega t)}$

Faraday's law $\nabla \times \vec{E} = \left(\nabla \mathrm{e}^{j(kx-\omega t)}\right) \times \vec{E_0} = jk \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = j\omega\vec{B}$

Since $\vec{B} = \frac{k}{\omega} \times \vec{E} = \frac{1}{c} k \times \vec{E}$:

$$\frac{F_{magnetic}}{F_{electrical}} \leq \frac{\mathrm{v}}{c}$$

电场磁场相互垂直，磁场大小等于电场大小除光速。

So, as long as a charge's velocity is much less than the speed of light, we can neglect the light's magnetic force compared to its electric force.